MATTHEW GOODBRED

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EDUCATION

Princeton University, Princeton, NJ *September 2024 – Expected 2029*
Ph.D. in Astrophysical Sciences

Dartmouth College, Hanover, NH *September 2019 – June 2023*
B.A., Physics with High Honors (*Magna Cum Laude*); Minor in Mathematics GPA: 3.94
Thesis: *The Structure and Rate of Relativistic Guide Field Magnetic Reconnection*
Phi Beta Kappa; Sigma Xi Scientific Research Honor Society

PUBLICATIONS

M. Goodbred and A. Spitkovsky, "Diocotron Modes in Pulsar Magnetospheres: Charge Diffusion and Implications for Radio Emission Variability," *The Astrophysical Journal* **1010**, 68 (2026). doi: 10.3847/1538-4357/ae9feb

M. Goodbred and Y.-H. Liu, "First-Principles Theory of the Relativistic Magnetic Reconnection Rate in Astrophysical Pair Plasmas," *Physical Review Letters* **129**, 265101 (2022). doi: 10.1103/PhysRevLett.129.265101

M. Goodbred, Y.-H. Liu, B. Chen, and X. Li, "The Relation Between the Energy Conversion Rate and Reconnection Rate in Petschek-Type Reconnection—Implications for Solar Flares," *Physics of Plasmas* **28**, 082103 (2021). doi:10.1063/5.0050557

RESEARCH EXPERIENCE

Graduate Research, Princeton University *September 2024 – Present*
Advisor: Anatoly Spitkovsky

- Ran 3D particle-in-cell simulations of pulsar magnetospheres on HPC clusters, resolving the disk–dome equilibrium and the onset of the diocotron instability.
- Showed the nonlinear diocotron state is a persistent $m = 1$ mode; derived a quasilinear diffusion coefficient for cross-field charge transport through the closed zone.
- Linked the instability to emission-beam perturbations and to observed nulling, periodic amplitude modulation, and drifting subpulses.

Magnetic Reconnection Group, Dartmouth College *June 2020 – June 2023*
Advisor: Yi-Hsin Liu

- Derived the first analytic theory of relativistic reconnection rates; confirmed with PIC simulations.
- Related energy conversion and reconnection rates in Petschek reconnection; applied to solar flares.

SELECTED TALKS AND PRESENTATIONS

A New Theory of Fast Magnetic Reconnection in Strongly Magnetized Plasmas
Oral, APS Division of Plasma Physics Meeting, Denver, CO *November 2023*

The Relation Between the Energy Conversion Rate and the Reconnection Rate in Petschek-Type Reconnection

- Oral, American Geophysical Union (AGU) Fall Meeting *December 2021*
- Invited talk, SoLFER Colloquium *October 2021*
- Poster, SHINE Heliophysics 2050 Conference *September 2021*

AWARDS AND FELLOWSHIPS

Gazzaniga Family Science Award, Dartmouth College *2023*
Awarded to the graduating senior with the strongest record of scientific research.

Francis W. Sears Physics and Astronomy Faculty Prize, Dartmouth College *2023*
Awarded for significant departmental contributions and unusual undergraduate achievement.

Wilder Undergraduate Research Fellowship, Dartmouth College *2022*

Francis L. Town Scientific Prize for Achievement in Physics, Dartmouth College *2021*

James O. Freedman Presidential Scholarship, Dartmouth College *2021*

TEACHING AND MENTORING

Graduate Student Mentor, Princeton University *2026 – Present*

Assistant in Instruction, AST/PHY 401: Cosmology, Princeton University *Spring 2026*

- Upper-level course on extragalactic astronomy and cosmology: galaxies, clusters and large-scale structure, dark matter and dark energy, the expanding and early Universe, the CMB, inflation, and structure formation.
- Led problem sessions and office hours, and graded problem sets and exams.

Learning Assistant, Physics 41: Electrodynamics, Dartmouth College *Winter 2022*

- Facilitated in-class problem sessions and weekly office hours; worked with the instructor on course design.

Physics and Computer Science Tutor, Dartmouth College *Winter 2021, Spring 2023*

SERVICE AND OUTREACH

Astro Coffee Host, Department of Astrophysical Sciences, Princeton University *2026 – Present*

Public Observing Volunteer, Princeton University *2024 – Present*

Physics/Astronomy Foreign Study Program, Cape Town, South Africa *January – March 2023*

- Conducted observations at the South African Astronomical Observatory; participated in outreach to local schools.

MEDIA COVERAGE

The Tiny Physics Behind Immense Cosmic Eruptions
Z. Savitsky, *Quanta Magazine* *May 2023*

Decoding Mega Magnetic Explosions Outside the Solar System
H. Barath, *Dartmouth News* *January 2023*

TECHNICAL SKILLS

Simulation: PIC, large-scale HPC/MPI workflows, N-body electrostatic codes
Languages and Tools: Python (NumPy/SciPy/h5py/Matplotlib), Jupyter, MATLAB, Mathematica, C, LaTeX, Git, Unix command line